

Module 2 Project Report

Linux, Networking, and Cloud Infrastructure Deployment on AWS EC2

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Module 2: Linux, Networking, and Cloud Infrastructure

Executive Summary:

This project demonstrates the deployment, configuration, and administration of a cloud-based Linux server using Amazon Web Services (AWS) Elastic Compute Cloud (EC2). The project involved launching an Ubuntu Server instance, establishing secure remote access using SSH, performing Linux system administration tasks, configuring user and group management, implementing network and firewall security controls, and deploying a web application using the Nginx web server. The successful completion of this project provided practical experience in cloud computing, Linux administration, networking fundamentals, system security, and web server deployment. The final outcome was a publicly accessible website hosted on an Ubuntu server running in the AWS cloud environment.

Introduction:

Cloud computing has become a fundamental component of modern IT infrastructure. Organizations increasingly rely on virtualized cloud resources to deploy scalable, secure, and reliable services. Linux operating systems play a significant role in cloud environments due to their stability, performance, and security.

This project was designed to provide hands-on experience in deploying and managing a Linux server within the AWS cloud platform. Through a series of practical exercises, essential concepts of Linux administration, networking, security, and web hosting were implemented and verified.

Project Objectives

The primary objectives of this project were:

- Deploy an Ubuntu Linux server on AWS EC2.
- Configure secure remote access using SSH.
- Perform Linux command-line administration tasks.
- Create and manage files and directories.
- Implement Linux file permissions.
- Configure users and groups.
- Monitor running processes and services.
- Configure networking and DNS services.
- Implement firewall security using UFW.
- Configure AWS Security Groups for external access.
- Install and configure the Nginx web server.
- Deploy and publish a website accessible through the public internet.

Technologies and Tools Used:

Technology	Purpose
AWS EC2	Cloud Infrastructure
Ubuntu Server LTS	Operating System
Linux CLI	System Administration
SSH	Secure Remote Access
PowerShell	Remote Connectivity
UFW Firewall	Linux Host Security
AWS Security Groups	Network Security
DNS Utilities	Name Resolution Testing
Nginx	Web Server Hosting
HTML	Website Development

Project Methodology

The project was completed through the following phases:

Phase 1: Cloud Infrastructure Deployment

- AWS EC2 service accessed.
- Ubuntu Server LTS selected.
- Security groups configured.
- SSH key pair generated.
- Virtual machine launched successfully.

Phase 2: Linux System Administration

- Remote SSH connectivity established.
- Linux commands executed.
- Directories and files created.
- File permissions configured.

Phase 3: User and Access Management

- Linux groups created.
- New users added.
- Group memberships assigned and verified.

Phase 4: Networking and Security

- Network settings inspected.
- DNS functionality tested.
- Firewall rules configured.
- AWS Security Group rules updated.

Phase 5: Web Server Deployment

- Nginx installed and configured.
- Web content developed.
- HTTP service enabled.
- Connectivity verified both locally and externally.

Implementation Details

AWS EC2 Instance Deployment

A virtual server instance was successfully deployed using AWS EC2. Ubuntu Server LTS was selected as the operating system image and a secure RSA-based key pair was used for SSH authentication.

The instance reached a healthy operational state with all system status checks passing successfully.

Figure 01: Ubuntu AMI Selection

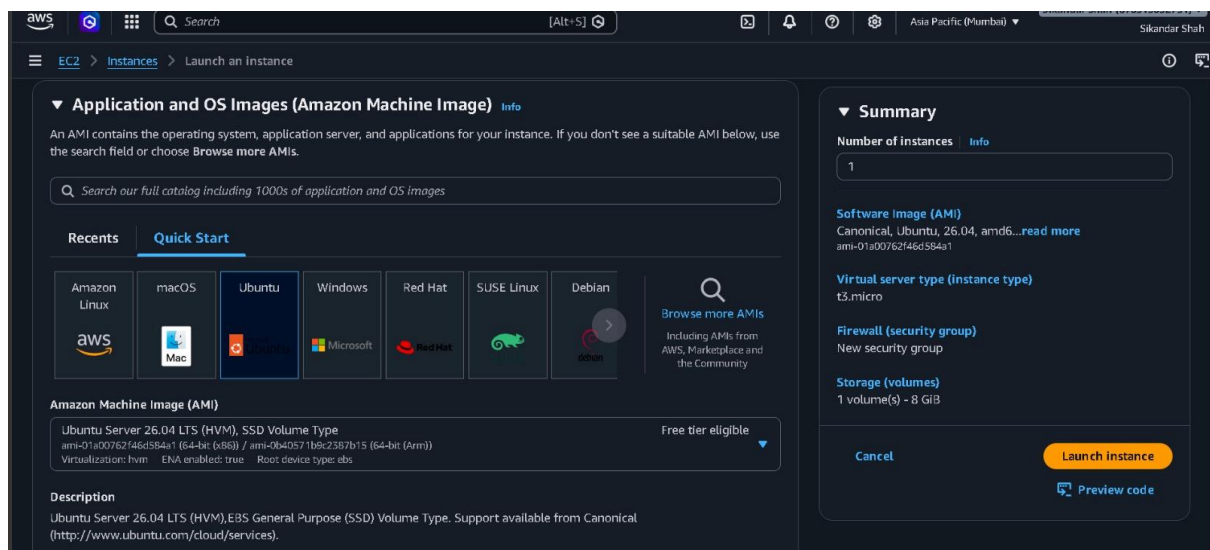


Figure 02: Instance Type Configuration

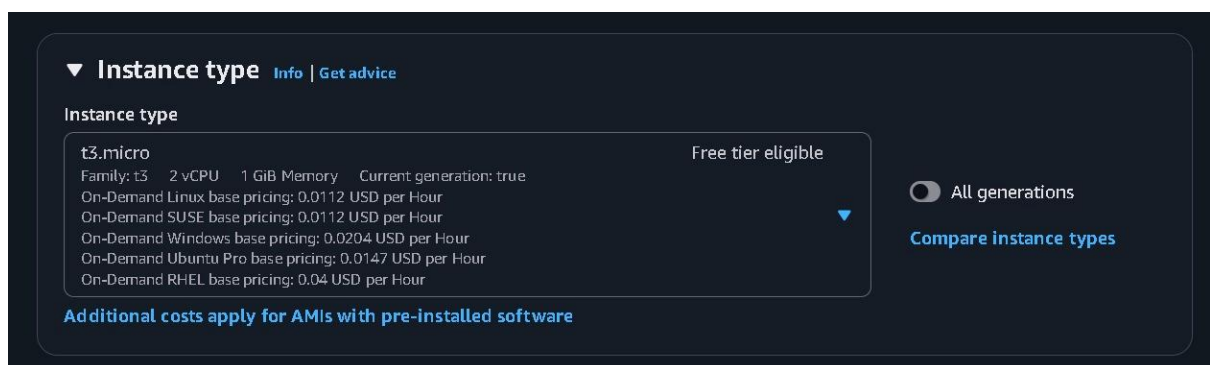


Figure 03: Key Pair Configuration

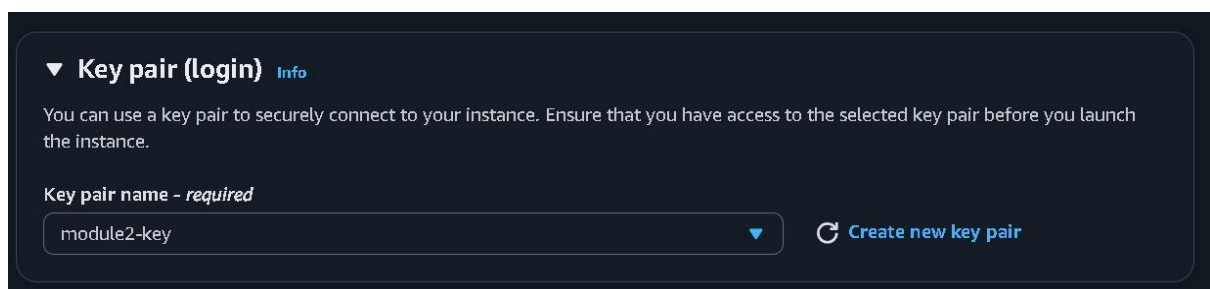


Figure 04: Security Group Configuration

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

module2-key ▼

↻ Create new key pair

Figure 05: Running EC2 Instance

Instances (3) Info

Last updated 1 minute ago

Connect ⓘ

Instance state ▼

Actions ▼

Launch instances ▼

Saved filter sets

Choose filter set

Filter instance by attribute or tag (case-sensitive)

☐	Name ⓘ	Instance ID	Instance state ▼	Instance type ▼	Status check	Availability Zone ▼	Public IPv4 DNS ▼	Publ
☐	Lab2-env	i-019df4e80fbad4992	Terminated ⓘ	t3.micro	–	ap-south-1b	–	–
☐	Lab2-env	i-0d762a4f1e7973d4c	Stopping ⓘ	t3.micro	–	ap-south-1b	ec2-65-2-173-92.ap-so...	65.2
☐	Module2-Linux-Server	i-00c06d28a55acf54a	Running ⓘ	t3.micro	3/3 checks passed	ap-south-1a	ec2-3-110-105-185.ap...	3.11

SSH Connectivity and Linux Administration

Secure shell (SSH) connectivity was established using the generated PEM key file. After logging into the remote Ubuntu server, several Linux commands were executed to collect system information and verify operating system functionality.

Commands executed included:

1. whoami
2. hostname
3. pwd
4. ls -la
5. cat /etc/os-release
6. uname -a

Figure 06: SSH Connection

```
PS C:\Users\HB Comp> cd "$HOME\Downloads"
PS C:\Users\HB Comp\Downloads> dir

Directory: C:\Users\HB Comp\Downloads

Mode                LastWriteTime         Length Name
----                -
-a----          9/20/2026   2:34 AM           519583 Copilot_20260920_023424.png
-a----          9/20/2026   2:56 AM           755328 Copilot_20260920_025605.png
-a----          9/20/2026   3:46 AM          2886560 MBSetup-2.2.exe
-a----          9/20/2026   3:15 AM          2364456 Module 1_ Cloud Computing Foundations.docx
-a----          9/21/2026   1:26 AM           1674 module2-key.pem

PS C:\Users\HB Comp\Downloads> ssh -i ".\module2-key.pem" ubuntu@3.110.10.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep 21 08:53:31 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   30.5% of 6.61GB Processes:      117
Memory usage: 26%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 172.31.

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-38-255:~$ |
```

Figure 07: Linux Command-Line Operations

```
Last login: Mon Sep 21 08:53:32 2026 from 160.191.158.154
ubuntu@ip-172-31-38-255:~$ whoami
ubuntu
ubuntu@ip-172-31-38-255:~$ hostname
ip-172-31-38-255
ubuntu@ip-172-31-38-255:~$ pwd
/home/ubuntu
ubuntu@ip-172-31-38-255:~$ ls -la
total 28
drwxr-x--- 4 ubuntu ubuntu 4096 Sep 21 08:53 .
drwxr-xr-x 3 root   root   4096 Sep 21 08:34 ..
-rw-r--r-- 1 ubuntu ubuntu 220  Feb 13  2026 .bash_logout
-rw-r--r-- 1 ubuntu ubuntu 3771 Feb 13  2026 .bashrc
drwx----- 2 ubuntu ubuntu 4096 Sep 21 08:53 .cache
-rw-r--r-- 1 ubuntu ubuntu 807  Feb 13  2026 .profile
drwx----- 2 ubuntu ubuntu 4096 Sep 21 08:34 .ssh
ubuntu@ip-172-31-38-255:~$ cat /etc/os-release
PRETTY_NAME="Ubuntu 26.04 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04 LTS (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
ubuntu@ip-172-31-38-255:~$ uname -a
Linux ip-172-31-38-255 7.0.0-1006-aws #6-Ubuntu SMP PREEMPT Tue May 26 12:04:34 UTC 2026 x86_64 GNU/Linux
ubuntu@ip-172-31-38-255:~$
```

File System Management

Directories were created for file organization, logging, and backup storage. Test files were generated, modified, copied, and verified using Linux file management utilities.

1. `mkdir -p ~/module2/files`
2. `mkdir -p ~/module2/logs`
3. `mkdir -p ~/module2/backup`
4. `touch ~/module2/files/test.txt`
5. `cp ~/module2/files/test.txt ~/module2/backup/`

Figure 08: Directory Management

```
ubuntu@ip-172-31-38-255:~$  
ubuntu@ip-172-31-38-255:~$ mkdir -p ~/module2/files  
ubuntu@ip-172-31-38-255:~$ mkdir -p ~/module2/logs  
ubuntu@ip-172-31-38-255:~$ mkdir -p ~/module2/backup  
ubuntu@ip-172-31-38-255:~$ ls -la  
total 32  
drwxr-x--- 5 ubuntu ubuntu 4096 Sep 21 09:05 .  
drwxr-xr-x 3 root root 4096 Sep 21 08:34 ..  
-rw-r--r-- 1 ubuntu ubuntu 220 Feb 13 2026 .bash_logout  
-rw-r--r-- 1 ubuntu ubuntu 3771 Feb 13 2026 .bashrc  
drwx----- 2 ubuntu ubuntu 4096 Sep 21 08:53 .cache  
-rw-r--r-- 1 ubuntu ubuntu 807 Feb 13 2026 .profile  
drwx----- 2 ubuntu ubuntu 4096 Sep 21 08:34 .ssh  
drwxrwxr-x 5 ubuntu ubuntu 4096 Sep 21 09:05 module2  
ubuntu@ip-172-31-38-255:~$ ls -la ~/module2  
total 20  
drwxrwxr-x 5 ubuntu ubuntu 4096 Sep 21 09:05 .  
drwxr-x--- 5 ubuntu ubuntu 4096 Sep 21 09:05 ..  
drwxrwxr-x 2 ubuntu ubuntu 4096 Sep 21 09:05 backup  
drwxrwxr-x 2 ubuntu ubuntu 4096 Sep 21 09:05 files  
drwxrwxr-x 2 ubuntu ubuntu 4096 Sep 21 09:05 logs  
ubuntu@ip-172-31-38-255:~$
```

Figure 09: File Operations

```
Last login: Mon Sep 21 09:01:05 2026 from 100.129.1.100.104  
ubuntu@ip-172-31-38-255:~$ touch ~/module2/files/test.txt  
ubuntu@ip-172-31-38-255:~$ echo "Module 2 Linux Practice" > ~/module2/files/test.txt  
ubuntu@ip-172-31-38-255:~$ cat ~/module2/files/test.txt  
Module 2 Linux Practice  
ubuntu@ip-172-31-38-255:~$ cp ~/module2/files/test.txt ~/module2/backup/  
ubuntu@ip-172-31-38-255:~$ find ~/module2 -type f  
/home/ubuntu/module2/files/test.txt  
/home/ubuntu/module2/backup/test.txt  
ubuntu@ip-172-31-38-255:~$
```


File Permission Management

Linux file permissions were examined and modified using the `chmod` command to implement secure access controls.

1. `chmod 640 ~/module2/files/test.txt`

This ensured controlled read and write permissions according to security best practices.

Figure 10: File Permissions

```
ubuntu@ip-172-31-38-255:~$  
ubuntu@ip-172-31-38-255:~$ ls -l ~/module2/files/test.txt  
-rw-r----- 1 ubuntu ubuntu 24 Sep 21 09:16 /home/ubuntu/module2/files/test.txt  
ubuntu@ip-172-31-38-255:~$ chmod 640 ~/module2/files/test.txt  
ubuntu@ip-172-31-38-255:~$ ls -l ~/module2/files/test.txt  
-rw-r----- 1 ubuntu ubuntu 24 Sep 21 09:16 /home/ubuntu/module2/files/test.txt  
ubuntu@ip-172-31-38-255:~$
```

User and Group Administration

A dedicated group and user account were created to demonstrate Linux user management and role-based access control.

1. `sudo groupadd module2users`
2. `sudo adduser module2student`
3. `sudo usermod -aG module2users module2student`

User configuration was validated using administrative verification commands.

Figure 11: Users and Groups

```
ubuntu@ip-172-31-38-255:~$  
ubuntu@ip-172-31-38-255:~$ sudo groupadd module2users  
ubuntu@ip-172-31-38-255:~$ getent group module2users  
module2users:x:1001:  
ubuntu@ip-172-31-38-255:~$ sudo adduser module2student  
New password:  
Retype new password:  
passwd: password updated successfully  
Changing the user information for module2student  
Enter the new value, or press ENTER for the default  
Full Name []: Sikandar  
Room Number []:  
Work Phone []:  
Home Phone []:  
Other []:  
Is the information correct? [Y/n] Y  
ubuntu@ip-172-31-38-255:~$ sudo usermod -aG module2users module2student  
ubuntu@ip-172-31-38-255:~$ id module2student  
uid=1002(module2student) gid=1002(module2student) groups=1002(module2student),100(users),1001(module2users)  
ubuntu@ip-172-31-38-255:~$ groups module2student  
module2student : module2student users module2users  
ubuntu@ip-172-31-38-255:~$
```

Process and Service Management

System processes and active services were monitored using standard Linux process management tools.

1. ps aux
2. ps -ef
3. top

The SSH daemon service was examined to verify operational status.

Figure 12: Process Monitoring

```
ubuntu@ip-172-31-38-255:~$ ps aux | grep sshd
root      1086  0.0  0.8 10712  7660 ?        Ss   08:35   0:00 sshd: /usr/sbin/sshd -D -o AuthorizedKeysCommand /usr/share/ec2-instance-connect/eic_run_authorized_key
s %u %f -o AuthorizedKeysCommandUser ec2-instance-connect [listener] 0 of 10-100 startups
root      1170  0.0  1.3 18116 12152 ?        Ss   08:53   0:00 sshd-session: ubuntu [priv]
ubuntu    1287  0.0  0.8 18416  7840 ?        S    08:53   0:00 sshd-session: ubuntu@pts/0
root      1328  0.0  1.2 18188 11940 ?        Ss   09:01   0:00 sshd-session: ubuntu [priv]
ubuntu    1386  0.0  0.8 18408  7848 ?        S    09:01   0:00 sshd-session: ubuntu@pts/1
root      1635  0.0  1.2 18112 11812 ?        Ss   09:15   0:00 sshd-session: ubuntu [priv]
ubuntu    1694  0.0  0.8 18412  7848 ?        S    09:15   0:00 sshd-session: ubuntu@pts/2
ubuntu    2010  0.0  0.2      1  2356 pts/2    S+   09:23   0:00 grep --color=auto sshd
ubuntu@ip-172-31-38-255:~$
```

Figure 13: SSH Service Verification

```
ubuntu@ip-172-31-38-255:~$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Drop-In: /usr/lib/systemd/system/ssh.service.d
            └─ec2-instance-connect.conf
   Active: active (running) since Mon 2026-09-21 08:35:00 UTC; 51min ago
   Invocation: 4853da959e04b5481ee64432d1c1f3e
   TriggeredBy: ● ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
   Main PID: 1086 (sshd)
   Tasks: 1 (limit: 627)
   Memory: 11.1M (peak: 12.4M)
   CPU: 163ms
   CGroup: /system.slice/ssh.service
           └─1086 "sshd: /usr/sbin/sshd -D -o AuthorizedKeysCommand /usr/share/ec2-instance-connect/eic_run_authorized_keys %u %f -o AuthorizedKeysCommandUser ec2-inst

Sep 21 08:35:00 ip-172-31-38-255 sshd[1086]: Server listening on 0.0.0.0 port 22.
Sep 21 08:35:00 ip-172-31-38-255 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Sep 21 08:35:00 ip-172-31-38-255 sshd[1086]: Server listening on :: port 22.
Sep 21 08:45:37 ip-172-31-38-255 sshd-session[1158]: Connection reset by authenticating user ubuntu 160.191.158.154 port 31077 [preauth]
Sep 21 08:53:30 ip-172-31-38-255 sshd-session[1170]: Accepted publickey for ubuntu from 160.191.158.154 port 31771 ssh2: RSA SHA256:alt/WLx4Zceb0mE9fdFqKUKHGHPZw/3HxTqvX
Sep 21 08:53:30 ip-172-31-38-255 sshd-session[1170]: pam_unix(sshd:session): session opened for user ubuntu(uid=1000) by ubuntu(uid=0)
Sep 21 09:01:52 ip-172-31-38-255 sshd-session[1320]: Accepted publickey for ubuntu from 160.191.158.154 port 31680 ssh2: RSA SHA256:alt/WLx4Zceb0mE9fdFqKUKHGHPZw/3HxTqvX
Sep 21 09:01:52 ip-172-31-38-255 sshd-session[1320]: pam_unix(sshd:session): session opened for user ubuntu(uid=1000) by ubuntu(uid=0)
Sep 21 09:15:18 ip-172-31-38-255 sshd-session[1635]: Accepted publickey for ubuntu from 160.191.158.154 port 31285 ssh2: RSA SHA256:alt/WLx4Zceb0mE9fdFqKUKHGHPZw/3HxTqvX
Sep 21 09:15:18 ip-172-31-38-255 sshd-session[1635]: pam_unix(sshd:session): session opened for user ubuntu(uid=1000) by ubuntu(uid=0)
ubuntu@ip-172-31-38-255:~$ systemctl is-active ssh
active
ubuntu@ip-172-31-38-255:~$
```

SSH Key Generation

A modern ED25519 SSH key pair was generated to strengthen authentication security.

1. `ssh-keygen -t ed25519 -C "module2-linux"`

The generated public key was displayed and verified successfully.

Figure 14: SSH Key Generation

```
ubuntu@ip-172-31-38-255:~$ ls -la ~/.ssh
total 12
drwx----- 2 ubuntu ubuntu 4096 Sep 21 08:34 .
drwxr-x--- 6 ubuntu ubuntu 4096 Sep 21 09:23 ..
-rw----- 1 ubuntu ubuntu 393 Sep 21 08:34 authorized_keys
ubuntu@ip-172-31-38-255:~$ ssh-keygen -t ed25519 -C "module2-linux"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/ubuntu/.ssh/id_ed25519):
Enter passphrase for "/home/ubuntu/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/ubuntu/.ssh/id_ed25519
Your public key has been saved in /home/ubuntu/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256: [REDACTED] module2-linux
The key's randomart image is:
+--[ED25519 256]--+
| o*o.o |
| oooooo  |
| +oo+ o + |
| . +.= + + |
|   o So + .+oo |
|     +..E= |
|     ...=oX |
|     . o. == |
|     .=. ..o |
+----[SHA256]-----+
ubuntu@ip-172-31-38-255:~$ ls -la ~/.ssh
total 20
drwx----- 2 ubuntu ubuntu 4096 Sep 21 09:28 .
drwxr-x--- 6 ubuntu ubuntu 4096 Sep 21 09:23 ..
-rw----- 1 ubuntu ubuntu 393 Sep 21 08:34 authorized_keys
-rw----- 1 ubuntu ubuntu 399 Sep 21 09:28 id_ed25519
-rw-r--r-- 1 ubuntu ubuntu 95 Sep 21 09:28 id_ed25519.pub
ubuntu@ip-172-31-38-255:~$ ^C
ubuntu@ip-172-31-38-255:~$ cat ~/.ssh/id_ed25519.pub
ssh-ed25519 AAAAC3NzaC1lbnRpdjEAAAAB [REDACTED] module2-linux
ubuntu@ip-172-31-38-255:~$
```

Networking Configuration

Network interface information, routing tables, and IP address configurations were examined using Linux networking commands.

1. ip addr
2. hostname -I
3. ip route

DNS resolution was tested successfully using domain lookup utilities.

1. dig example.com
2. nslookup example.com

Figure 15: Network Configuration

```
ubuntu@ip-172-31-38-255:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 02:ff:ca:39:9b:31 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname enx02ffca399b31
    inet 172.31.38.255/20 metric 100 brd 172.31.47.255 scope global dynamic ens5
        valid_lft 2003sec preferred_lft 2003sec
    inet6 fe80::ff:caff:fe39:9b31/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
ubuntu@ip-172-31-38-255:~$ hostname -I
172.31.38.255
ubuntu@ip-172-31-38-255:~$ ip route
default via 172.31.32.1 dev ens5 proto dhcp src 172.31.38.255 metric 100
172.31.0.2 via 172.31.32.1 dev ens5 proto dhcp src 172.31.38.255 metric 100
172.31.32.0/20 dev ens5 proto kernel scope link src 172.31.38.255 metric 100
172.31.32.1 dev ens5 proto dhcp scope link src 172.31.38.255 metric 100
ubuntu@ip-172-31-38-255:~$ |
```

Figure 16: DNS Verification

```
ubuntu@ip-172-31-38-255:~$ dig example.com
;; <<>> DiG 9.20.24-lubuntu0.3-Ubuntu <<>> example.com
;; global options: +cmd
;; Got answer:
;;->HEADER<<- opcode: QUERY, status: NOERROR, id: 10603
;; flags: qr rd ra; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
;; EDNS: version: 0, flags:; udp: 65494
;; QUESTION SECTION:
;example.com.                IN      A

;; ANSWER SECTION:
example.com.                 101     IN      A      172.66.147.243
example.com.                 101     IN      A      104.20.23.154

;; Query time: 1 msec
;; SERVER: 127.0.0.53#53(127.0.0.53) (UDP)
;; WHEN: Mon Sep 21 09:33:18 UTC 2026
;; MSG SIZE rcvd: 72

ubuntu@ip-172-31-38-255:~$ nslookup example.com
Server:      127.0.0.53
Address:     127.0.0.53#53

Non-authoritative answer:
Name:   example.com
Address: 172.66.147.243
Name:   example.com
Address: 104.20.23.154
Name:   example.com
Address: 2606:4700:10::ac42:93f3
Name:   example.com
Address: 2606:4700:10::6814:179a
ubuntu@ip-172-31-38-255:~$ |
```

Firewall and Security Configuration

Host-level firewall rules were implemented using Uncomplicated Firewall (UFW).

1. `sudo ufw allow 22/tcp`
2. `sudo ufw allow 80/tcp`
3. `sudo ufw enable`

The firewall was configured to securely permit SSH and HTTP traffic.

Figure 17: Port Verification

```
ubuntu@ip-172-31-38-255:~$ sudo ss -tulpn
Netid: State Recv-Q Send-Q Local Address:Port Peer Address:Port Process
udp UNCONN 0 0 127.0.0.54:53 0.0.0.0:* users:({systemd-resolve,pid=197,fd=18})
udp UNCONN 0 0 127.0.0.53%lo:53 0.0.0.0:* users:({systemd-resolve,pid=197,fd=16})
udp UNCONN 0 0 172.31.38.255%ens5:68 0.0.0.0:* users:({systemd-networkd,pid=593,fd=36})
udp UNCONN 0 0 127.0.0.1:323 0.0.0.0:* users:({chronyd,pid=732,fd=4})
udp UNCONN 0 0 127.0.0.53%lo:53 0.0.0.0:* users:({chronyd,pid=732,fd=5})
tcp LISTEN 0 4096 127.0.0.53%lo:53 0.0.0.0:* users:({systemd-resolve,pid=197,fd=17})
tcp LISTEN 0 4096 0.0.0.0:22 0.0.0.0:* users:({sshd,pid=1006,fd=3},{systemd,pid=1,fd=252})
tcp LISTEN 0 4096 127.0.0.54:53 0.0.0.0:* users:({systemd-resolve,pid=197,fd=19})
tcp LISTEN 0 4096 0.0.0.0:22 0.0.0.0:* users:({sshd,pid=1006,fd=4},{systemd,pid=1,fd=253})
ubuntu@ip-172-31-38-255:~$ sudo ss -tulpn | grep :22
tcp LISTEN 0 4096 0.0.0.0:22 0.0.0.0:* users:({sshd,pid=1006,fd=3},{systemd,pid=1,fd=252})
tcp LISTEN 0 4096 0.0.0.0:22 0.0.0.0:* users:({sshd,pid=1006,fd=4},{systemd,pid=1,fd=253})
ubuntu@ip-172-31-38-255:~$
```

Figure 18: UFW Firewall Configuration

```
ubuntu@ip-172-31-38-255:~$ sudo ufw status
Status: inactive
ubuntu@ip-172-31-38-255:~$ sudo ufw allow 22/tcp
Rules updated
Rules updated (v6)
ubuntu@ip-172-31-38-255:~$ sudo ufw allow 80/tcp
Rules updated
Rules updated (v6)
ubuntu@ip-172-31-38-255:~$ sudo ufw enable
Command may disrupt existing ssh connections. Proceed with operation (y|n)? y
Firewall is active and enabled on system startup
ubuntu@ip-172-31-38-255:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
80/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)

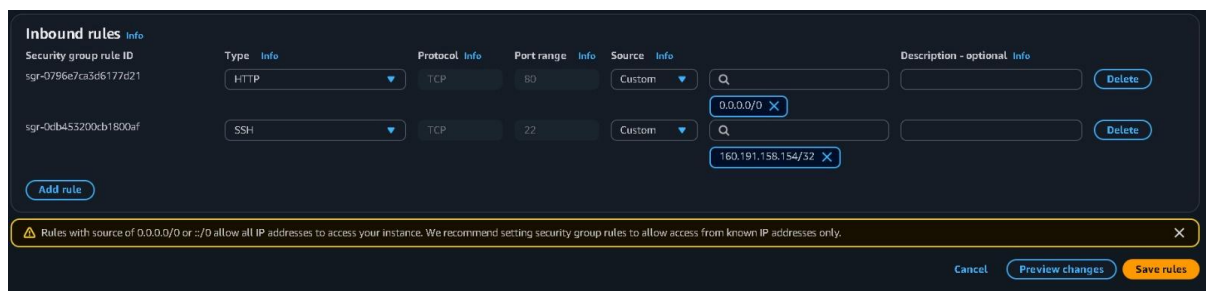
ubuntu@ip-172-31-38-255:~$
```

AWS Security Group Configuration

AWS Security Groups were updated to allow inbound HTTP traffic on port 80, enabling public access to the hosted website.

Service	Protocol	Port
SSH	TCP	22
HTTP	TCP	80

Figure 19: AWS Security Group Rules



Nginx Web Server Deployment

The Nginx web server was installed and activated successfully.

1. `sudo apt update`
2. `sudo apt install nginx -y`

Service functionality was validated through system service checks.

1. `sudo systemctl status nginx`

Figure 20: Nginx Installation

```
ubuntu@ip-172-31-38-255:~$ sudo apt update
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
170 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-38-255:~$ sudo apt install nginx -y
Installing:
  nginx

Installing dependencies:
  nginx-common

Suggested packages:
  fcgiwrap nginx-doc ssl-cert

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 170
  Download size: 666 kB
  Space needed: 1880 kB / 4641 MB available

Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 nginx-common all 1.28.3-2ubuntu1.11 [37.9 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64v3 nginx amd64 1.28.3-2ubuntu1.11 [628 kB]
Fetched 666 kB in 0s (15.9 MB/s)
Preconfiguring packages ...
Selecting previously unselected package nginx-common.
(Reading database ... 85303 files and directories currently installed.)
Preparing to unpack .../nginx-common_1.28.3-2ubuntu1.11_all.deb ...
Unpacking nginx-common (1.28.3-2ubuntu1.11) ...
Selecting previously unselected package nginx.
Preparing to unpack .../nginx_1.28.3-2ubuntu1.11_amd64v3.deb ...
Unpacking nginx (1.28.3-2ubuntu1.11) ...
Setting up nginx-common (1.28.3-2ubuntu1.11) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/nginx.service' -> '/usr/lib/systemd/system/nginx.service'.
Setting up nginx (1.28.3-2ubuntu1.11) ...
* Upgrading binary nginx
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for ufw (0.36.2-9build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.
```

Figure 21: HTTP Port Verification

```
ubuntu@ip-172-31-38-255:~$ sudo ss -tulpn | grep :80
tcp    LISTEN 0      511             0.0.0.0:*        users:(("nginx",pid=3316,fd=5),("nginx",pid=3315,fd=5),("nginx",pid=3312,fd=5))
tcp    LISTEN 0      511             [::]:80         [::]:*        users:(("nginx",pid=3316,fd=6),("nginx",pid=3315,fd=6),("nginx",pid=3312,fd=6))
ubuntu@ip-172-31-38-255:~$
```

Website Deployment

A custom HTML webpage was created.

<h1>Module 2 - Linux, Networking & Cloud Infrastructure</h1>

<p>Submitted by: Sikandar Shah (Founder & CEO of Zero Trust Pro) </p>

<p>Linux server deployed successfully. </p>

<p>CodoMax Digital Solutions</p>

Nginx was restarted:

1. sudo systemctl restart nginx

Figure 22: Nginx Service Running

```
ubuntu@ip-172-31-38-255:~$ sudo systemctl restart nginx
ubuntu@ip-172-31-38-255:~$ sudo systemctl is-active nginx
active
ubuntu@ip-172-31-38-255:~$ |
```

Connectivity Testing

Local Testing

1. curl http://localhost
2. curl -I http://localhost

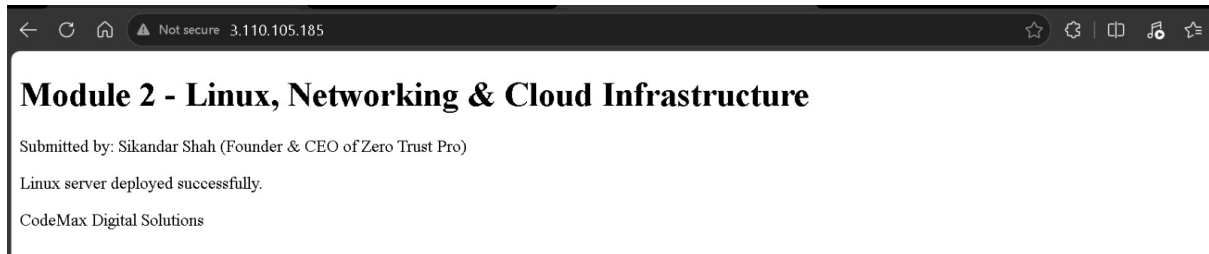
Figure 23: Local Connectivity Test

```
ubuntu@ip-172-31-38-255:~$ curl http://localhost
<!DOCTYPE html>
<html>
<head>
  <title>Module 2 Linux Server</title>
</head>
<body>
  <h1>Module 2 - Linux, Networking & Cloud Infrastructure</h1>
  <p> Submitted by: Sikandar Shah (Founder & CEO of Zero Trust Pro) </p>
  <p>Linux server deployed successfully.</p>
  <p>CodeMax Digital Solutions</p>
</body>
</html>
ubuntu@ip-172-31-38-255:~$ curl -I http://localhost
HTTP/1.1 200 OK
Server: nginx/1.28.3 (Ubuntu)
Date: Mon, 21 Sep 2026 09:49:32 GMT
Content-Type: text/html
Content-Length: 326
Last-Modified: Mon, 21 Sep 2026 09:47:24 GMT
Connection: keep-alive
ETag: "6ab0fd2c-146"
Accept-Ranges: bytes
ubuntu@ip-172-31-38-255:~$ |
```


Public Website Testing

The website was successfully accessed through the EC2 Public IPv4 address using a web browser.

Figure 24: Final Website Verification



Results and Verification

The project achieved all required objectives:

- ❖ Ubuntu EC2 instance deployed successfully
- ❖ SSH connectivity established
- ❖ Linux commands executed successfully
- ❖ Users and groups configured
- ❖ Firewall configured
- ❖ DNS functionality tested
- ❖ Nginx installed and operating
- ❖ Website hosted successfully
- ❖ Public internet accessibility verified

Learning Outcomes

Through this project, I learned:

- ❖ AWS EC2 deployment and management
- ❖ SSH authentication and key management
- ❖ Linux command-line administration
- ❖ User and group management
- ❖ Linux file permissions
- ❖ Networking and DNS troubleshooting
- ❖ UFW firewall configuration
- ❖ AWS Security Group management
- ❖ Nginx installation and management
- ❖ Web server deployment in a cloud environment

Challenges Faced

- ❖ Understanding AWS networking concepts.
- ❖ Managing SSH key permissions.
- ❖ Configuring firewall access correctly.
- ❖ Verifying web server accessibility through public IP.
- ❖ Troubleshooting service and network-related issues.

These challenges were resolved through systematic testing and verification.

Conclusion

This Module 2 project successfully demonstrated the deployment and administration of a Linux-based cloud server using AWS EC2. The project included Linux management, networking, security configuration, and web server deployment. The final result was a publicly accessible website hosted on an Ubuntu server using Nginx, validating the successful implementation of cloud infrastructure and Linux administration concepts.